

Weight of Raju = 68 kg

Weight of Vinay = 46 kg

Then, from (ii)

Weight of Pratap = $144 - 68 = 76$ kg

And from (i)

Weight of Arun = $204 - 124 = 80$ kg

Arun > Suraj > Pratap > Raju > Vinay

Second highest weight gainer = Suraj

Total weight of (Arun + Suraj + Vinay + Raju)
 $= (68 \times 4) = 272$ kg

Weight of Suraj = 78 kg

Weight of Raju = 68 kg

Weight of Vinay = 46 kg

Weight of Pratap is not known.

Only statement I alone was sufficient to answer the question.

- 165.** Average marks of Subodh in History Geography and Chemistry = 75

Total marks of H + G + C = $75 \times 3 = 225$... (i)

H + G + P = $78 \times 3 = 234$... (ii)

From (i) and (ii)

Marks of Subodh in Physics = $234 - 225 = 9$

- 166.** Let the three positive numbers in increasing order be a , b and c and the average of these numbers be A .

Then, $\frac{a+b+c}{3} = A$... (i)

Given $c - \frac{A}{3} = 8$

or, $c - \frac{a+b+c}{9} = 8$... (ii)

Also, given $\frac{b+a}{2} = 8$... (iii)

$\Rightarrow a + b = 16$

Putting the value of $(a + b)$ in equation (ii), we get

$$c - \left(\frac{16+c}{9} \right) = 8$$

or, $9c - 16 - c = 72$

or, $8c = 72 + 16 = 88$

or, $8z = 88$

$\therefore c = 11$

$\therefore$ Highest number = 11

- 167.** Average sales per day for six days of the week = ₹ 15,640/-

Total sales of six days of the week = $15640 \times 6 = ₹ 93,840/-$

Average sales of Tuesday to Saturday = ₹ 14,124/-

Total sales from Tuesday to Saturday = $14,124 \times 5$
 $= ₹ 70,620/-$

$\therefore$ Sales on Sunday = $(₹ 93,840 - 70,620) = ₹ 23,220/-$

- 168.** Average temperature of first four days = 25°C

Total temperature of first four days = $25^\circ \times 4 = 100^\circ\text{C}$

Average temperature last four days = 25.5°

Total temperature of four days = $25.5^\circ \times 4 = 102^\circ\text{C}$

Total temperature of whole week = $25.2 \times 7 = 176.4^\circ\text{C}$

$\therefore$ Temperature of the 4th day = $100^\circ + 102^\circ - 176.4^\circ = 25.6^\circ\text{C}$

- 169.** Sum of numbers = $205 + 302 + 108 + 403 + 202 = 1220$

Required average = $\frac{1220}{5} = 244$

- 170.** Average monthly income of P and Q = ₹ 6,000/-

Average monthly income of Q and R = ₹ 5,250/-

Average monthly income of P and R = ₹ 5,500/-

Total income of P + Q = $2 \times 6,000 = ₹ 12,000/-$... (i)

Total income of Q + R = $2 \times 5,250 = ₹ 10,500/-$... (ii)

Total income of R + P = $2 \times 5,500 = ₹ 11,000/-$... (iii)

On adding equation (i) (ii) and (iii), we get

$2(P + Q + R) = 12,000 + 10,500 + 11,000$

$\Rightarrow P + Q + R = \frac{33500}{2}$

= ₹ 16,750/- ... (iv)

By equation (iv) - (ii)

P's monthly income

= ₹ $(16,750 - 10,500) = ₹ 6,250/-$

- 171.** Average of 6 numbers = 7

Sum of 6 numbers = $6 \times 7 = 42$

Average of three numbers = 5

Sum of three numbers = $5 \times 3 = 15$

$\therefore$ Sum of the remaining three numbers

= $42 - 15 = 27$

$\therefore$ Required average = $\frac{27}{3} = 9$

- 172.** Let the number of boys and girls in the class are x and y .

According to given information

$30x + 20y = 23.25(x + y)$

$30x + 20y = 23.25x + 23.25y$

$30x - 23.25x = 23.25y - 20y$

$6.75x = 3.25y$

or, $\frac{x}{y} = \frac{3.25}{6.75} = \frac{13}{27}$

Hence, possible number of boys and girls 13 and 27 respectively.

- 173.** Average of a , b and c = 11

Total of a , b and c = 33 ... (i)

Similarly, Average of c , d , and e = 17

Sum of $c + d + e = 3 \times 17 = 51$ (ii)

Average of e and f is 22

Sum of $e + f = 2 \times 22 = 44$... (iii)

Average of e and c is 17

Sum of $e + c = 2 \times 17 = 34$... (iv)

By equations (i) + (ii) + (iii) - (iv)

$a + b + c + c + d + e + e + f - e - c$

= $33 + 51 + 44 - 34$

= $128 - 34 = 94$

$\therefore$ Required average = $\frac{94}{6}$